



Chapter 1

Introduction

Over the years, technology has made human life easier through continuous improvement and digital transformations. Every day, new technologies are developed and adopted across industries, reshaping how organizations or businesses operate, communicate, and deliver services. According to David et al. (2023), local governments' adoption of digital technologies has become a critical strategy for improving service delivery, transparency, and administrative efficiency, even though many public institutions continue to face fragmented infrastructures, limited technical expertise, and resistance to organizational change. Szedmák et al. (2025) also showed that implementing centralized digital platforms across more than 3,000 municipalities in Hungary resulted in faster case processing, shorter administrative lead times, and improved accountability in public service delivery. (It should be exact to the point and aligns with the rrl's)

The impact of digital transformation on governance has been particularly significant in the way public agencies manage records, process transactions, and respond to citizen concerns. Bhadouria et al. (2021) found that automating complaint workflows and providing real-time visibility into complaint status and departmental performance contribute to more efficient grievance management, increased accountability, and time-saving benefits for both the public and service agencies. In the Philippines, Andaya, Orlina, and Ilustre (2025) noted that while digital governance has improved public service responsiveness and accessibility, persistent barriers such as low digital literacy, inadequate ICT infrastructure, financial constraints, and institutional resistance continue to limit the full potential of e-government initiatives, particularly at the local level. (Talks about challenge)



In the field of housing governance, the need for integrated digital systems is especially pressing. Housing authorities such as the Talisay City Housing Authority are tasked with identifying and monitoring informal settler families, processing housing applications, managing required documentation, coordinating housing project implementation, and addressing housing-related concerns and complaints from residents.

The proponents aim to develop and design an Integrated Housing Services and Monitoring System for the Talisay City Housing Authority, a web-based platform that consolidates informal settler family monitoring, housing application management, document tracking, housing project coordination, complaint and case management, and analytics-driven reporting into an integrated platform to address the operational gaps of the Housing Authority and enhance the delivery of housing services across Talisay City.

1.1 Background of the Study

The City of Talisay is a fourth-class city in the province of Negros Occidental, Philippines, covering 27 barangays. Based on the 2015 Philippine Statistics Authority (PSA) census, the city has a population of 102,214, with an annual growth rate of 0.89%. Due to a growing population, the city faces significant housing challenges, with informal settlements being a consistent issue. According to Inquirer News (2022), Negros Occidental province has documented over 166,000 informal settler families, with significant concentrations in urban and peri-urban areas, including Talisay City.

The Talisay City Housing Authority is the local government agency responsible for managing housing programs and addressing housing needs within the city. The Authority identifies and monitors informal settler families across the 27 barangays, processes housing applications from eligible families, manages required documentation, coordinates the



implementation of housing projects and unit assignments, and addresses housing-related concerns. Currently, these critical operations are managed through manual, disconnected processes, with administrative and technical staff coordinating across multiple data sources and paper-based systems without integration across functions.

Talisay City's participation in the Negros Occidental Impact 2025 (NOI25) program further underscores the Housing Authority's critical role in municipal housing governance. The NOI25 program aims to construct 10,000 disaster-resilient homes targeting informal settlers and low-income families across Negros Occidental province. This initiative requires effective local-level coordination, beneficiary identification, application processing, and progress monitoring.

Inefficient housing governance processes and fragmented administrative systems are persistent challenges in government field offices. Despite growing recognition of the need for digital transformation in public administration, a centralized system specifically designed for housing regulatory operations remains absent. Existing systems, both internationally and locally, tend to address only isolated functions. Uriawan et al. (2023) developed the e-Perumahan web-based housing management system, which centralized housing data and improved stakeholder communication, but focused primarily on resident services rather than regulatory compliance.

Dela Cerna et al. (2023) implemented a modular office automation system for a Philippine barangay that covered complaint filing and document management, but was not designed for the scale or regulatory requirements of a national housing agency's field office. Dela Rosa (2022) developed a web-based case management information system for the National Labor Relations Commission that enabled real-time case tracking, yet it addressed labor dispute resolution rather than housing operations. Nerona et al. (2025) demonstrated that even simple digital systems could reduce data retrieval times and dramatically improve



data management quality in a barangay setting, but their Excel VBA-based approach was not scalable to the multi-functional demands of a regional housing office.

The proposed Integrated Housing Services and Monitoring System seeks to consolidate the Talisay City Housing Authority's existing operations into a unified web-based platform to streamline operations, improve data accessibility, and enable evidence-based decision-making.

1.2 Objectives of the Study

General Objectives:

The purpose of the study is to design, develop, and evaluate an Integrated Housing Services and Monitoring System that digitizes and integrates informal settler family monitoring, housing application management, document tracking, housing project and unit monitoring, case management, and reporting functions for the Talisay City Housing Authority.

Specifically, this study aims to:

1. To develop an Informal Settler Family (ISF) Monitoring module that records demographic information and generates statistics on ISF distribution across barangays in Talisay City.
2. To develop a Housing Application Management module that records and tracks applications submitted by informal settler families for housing programs.
3. To develop a Document Management and Tracking module that monitors application requirements and records the status of submitted documents.
4. To develop a Housing Project and Unit Monitoring module that tracks housing project information and housing unit availability for beneficiary allocation.



5. To develop a Case Management module that records, categorizes, and tracks housing-related concerns and complaints based on positive, neutral, or negative classifications.
6. To develop an Analytics and Reporting Dashboard that summarizes ISF statistics, application status, housing unit availability, and case records to support operational monitoring and decision-making.
7. To evaluate the system's usability and effectiveness through user acceptance testing and performance evaluation.

1. All about title and system functionality

1.1

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1.3

2. Test functionality of the system

3. Standard evaluation criteria (like ISO

4. User Guide

1.3 Significance of the Study

This study is significant as it proposes the development of an Integrated Housing Services and Monitoring System tailored for the Talisay City Housing Authority. The study will be beneficial to the following:

Talisay City Housing Authority, The system will provide centralized access to housing-related data through analytics and reporting dashboards. This enables the Housing Authority to efficiently monitor Informal Settler Family (ISF) records, housing applications, housing unit availability, and case records, supporting more informed decision-making and improved management of housing programs.



Administrative and Technical Staff of the Talisay City Housing Authority. The system aims to reduce reliance on manual processes and fragmented record-keeping. By digitizing ISF monitoring, application tracking, document management, and case recording, staff can more easily organize records, retrieve information, and monitor the status of housing applications and concerns.

Residents and Housing Applicants, the system will improve transparency and organization in the handling of housing applications and concerns. By ensuring that applications and cases are properly recorded and monitored, the Housing Authority can respond to inquiries and housing-related concerns more efficiently, thereby improving service delivery for residents.

Future Researchers, The findings of this study may serve as a reference for researchers who intend to develop or improve systems related to housing management, local government service automation, and digital governance platforms.

1.4 Scope and Limitation

This study focuses on developing an Integrated Housing Services and Monitoring System for the Talisay City Housing Authority. The system is a web-based platform designed to digitize and integrate the office's core operations, including Informal Settler Family monitoring, housing application management, document tracking, housing project and unit monitoring, case management, and operational reporting. The system will provide user-friendly interfaces for authorized staff of the Talisay City Housing Authority, enabling them to record ISF data, manage housing applications, track document submissions, monitor housing unit availability, record housing-related concerns, and generate summary reports.

Regarding document management, the system does not perform document completeness verification at intake, as the Housing Authority's existing intake procedure



already ensures that only complete document packages are accepted prior to system entry.

The Document Management and Tracking Module is therefore scoped to record received documents, monitor their processing stage, and maintain a digital archive linked to each applicant's record.

While the system offers several features to improve the office's operations, it also has limitations to consider. The system is limited to the operations and workflows of the Talisay City Housing Authority only and does not extend to other housing agencies or government offices. The accuracy of the tracking, monitoring, and reporting tools depends heavily on the quality and completeness of the data entered by office staff, and incomplete or incorrect entries could affect the reliability of generated reports and analytics. The system is developed strictly for academic purposes and is not intended for live deployment without a separate agreement. It does not include a public-facing portal or mobile application, integrate with other government systems, or incorporate artificial intelligence or machine learning features. All data encoding and management is handled internally by authorized office personnel.

1.5 Definition of Terms

The following key terms have been operationally defined in relation to their specific usage in this study. These definitions aim to provide clarity and ensure a consistent understanding of concepts central to the study.

Analytics Dashboard. In this study, it refers to a visual interface within the system that presents summarized operational data, including ISF statistics, housing application status, housing unit availability, and recorded case information.



Audit Trail. In this study, it refers to a digital record that logs actions performed within the system, including updates to ISF records, changes to application status, document updates, and case entries, for accountability and traceability.

Case Management Module. In this study, it refers to a system component that records, categorizes, and tracks housing-related concerns or complaints, allowing staff to monitor their status and resolution.

Document Management and Tracking Module. In this study, it refers to a system component that records and monitors application requirements and document submissions for housing applicants.

Housing Application Management Module, In this study, it refers to a system component that records and tracks applications submitted by Informal Settler Families for housing programs.

Housing Project and Unit Monitoring Module. In this study, it refers to a system component that records housing project information and monitors the availability and assignment of housing units for beneficiaries.

Informal Settler Family (ISF). In this study, it refers to a household residing in an area without formal legal ownership of the land or housing they occupy.

Informal Settler Family (ISF) Monitoring Module. In this study, it refers to a system component that centralizes the recording of ISF demographic information, housing status, and barangay-level distribution data, enabling the Housing Authority to conduct needs assessment and program targeting across all 27 barangays of Talisay City.

Integrated Housing Services and Monitoring System. In this study, it refers to a web-based system designed to integrate Informal Settler Family monitoring, housing



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application management, document tracking, housing project and unit monitoring, case management, and analytics reporting for the Talisay City Housing Authority.

Role-Based Access Control. In this study, it refers to a system security feature that restricts access and permissions according to the user's assigned role within the system.



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